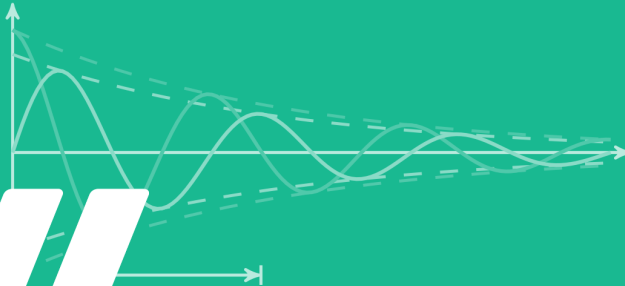


GET it digital

Modul 8: Schaltungen variabler Frequenz



Stand: 11. Februar 2026



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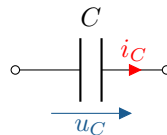
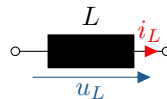
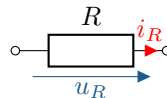
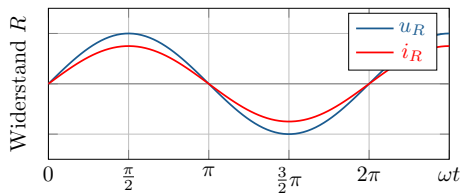
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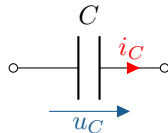
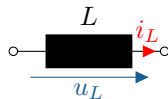
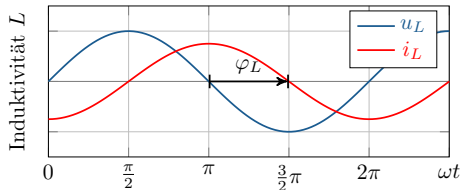
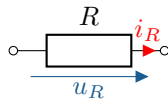
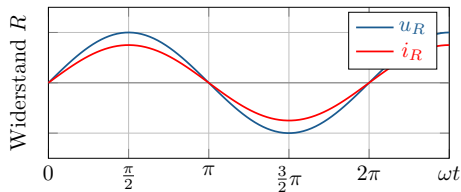
The work is available online at:

<https://getitdigital.uni-wuppertal.de/module/modul-8-schaltungen-variabler-frequenz>

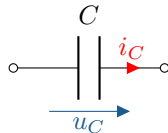
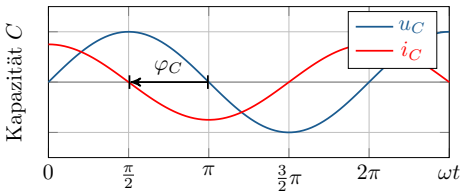
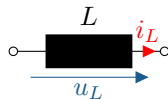
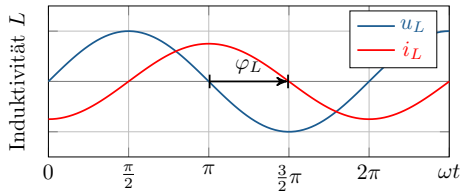
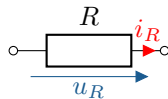
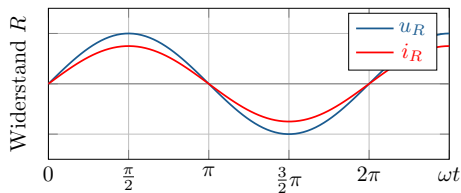
Frequency dependence of electrical components



Frequency dependence of electrical components

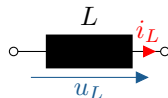


Frequency dependence of electrical components



Inductance

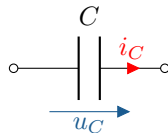
- ▶ Impedanz: $\underline{Z}_L = j\omega L$
- ▶ DE: $u_L = L \cdot \frac{d}{dt} i_L$
- ▶ Energy in the magnetic field
- ▶ Steady currents (no jumps)



Coil

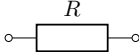
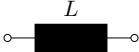
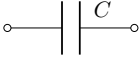
Capacity

- ▶ Impedance: $\underline{Z}_C = -j\frac{1}{\omega C}$
- ▶ DE: $i_C = C \cdot \frac{d}{dt} u_C$
- ▶ Energy in the electric field
- ▶ Voltages steady (no jumps)



Capacitor

Comparison of linear two-pole elements R , L and C

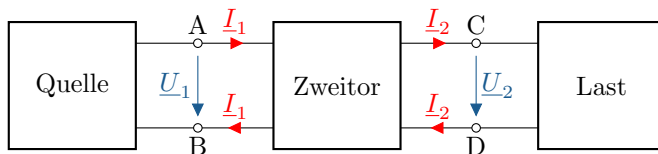
Size	Generally	El. resistance	Inductance	Capacity
Symbol				
Unit	[Form.s.] = Unit	$[R] = \Omega$ (Ohm)	$[L] = \text{H}$ (Henry)	$[C] = \text{F}$ (Farad)
Time domain	$\frac{d}{dt}$ or $\int dt$	$u_R = R \cdot i_R$	$u_L = L \cdot \frac{d}{dt} i_L$	$i_C = C \cdot \frac{d}{dt} u_C$
frequency range	$j\omega$ or $\frac{1}{j\omega}$	$\underline{U}_R = R \cdot \underline{I}_R$	$\underline{U}_L = j\omega L \cdot \underline{I}_L$	$\underline{I}_C = j\omega C \cdot \underline{U}_C$
Impedance	$\underline{Z} = \frac{\underline{U}}{\underline{I}}$	$\underline{Z}_R = R$	$\underline{Z}_L = j\omega L$	$\underline{Z}_C = -j\frac{1}{\omega C}$
Active part	$R = \text{Re}\{\underline{Z}\}$	$R = \frac{U}{I}$	0	0
Blind part	$X = \text{Im}\{\underline{Z}\}$	0	$X_L = \omega L$	$X_C = -\frac{1}{\omega C}$

Learning objectives: Frequency response, amplitude response, phase response

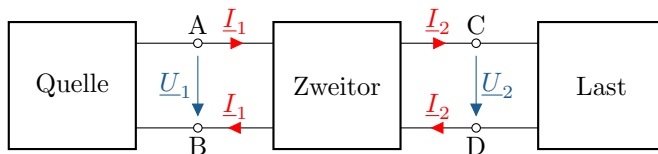
Students learn:

- ▶ Determine frequency responses algebraically based on circuit topology
- ▶ Understand how simple high-pass and low-pass filters work
- ▶ Analyse the limiting behaviour of four-poles based on their frequency responses

Analysis of frequency-variable behaviour of electrical networks

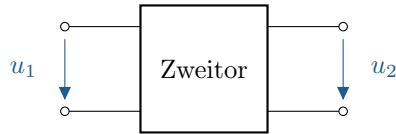


Analysis of frequency-variable behaviour of electrical networks



What is the transmission behavior like?

Analysis of frequency-variable behaviour of electrical networks (Two-port networks)



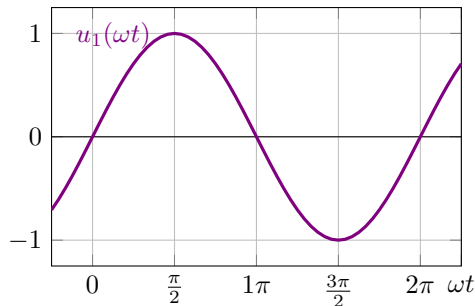
Linear and time-invariant (LTI):

Analysis of frequency-variable behaviour of electrical networks (Two-port networks)



Linear and time-invariant (LTI):

- Sine waveform at the input $u_1(\omega t)$

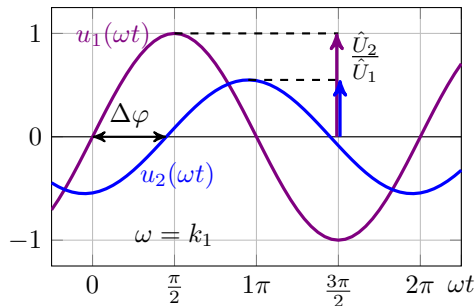


Analysis of frequency-variable behaviour of electrical networks (Two-port networks)



Linear and time-invariant (LTI):

- ▶ Sine waveform at the input $u_1(\omega t)$
- ▶ → Sine waveform at the output $u_2(\omega t)$
- ▶ → Frequency equal $\omega_2 = \omega_1$

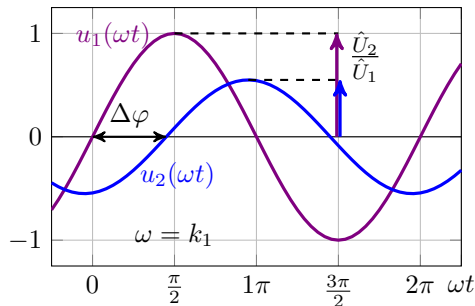


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- ▶ → Amplitude ratio $\frac{\hat{U}_2}{\hat{U}_1} = ?$

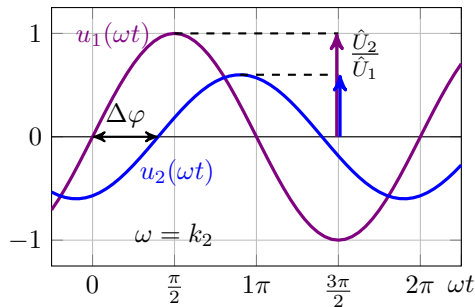


Analysis of frequency-variable behaviour of electrical networks (Two-port networks)

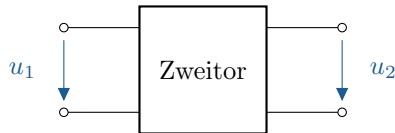


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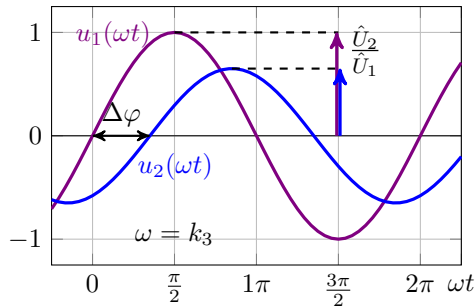


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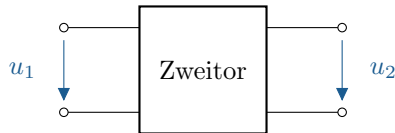


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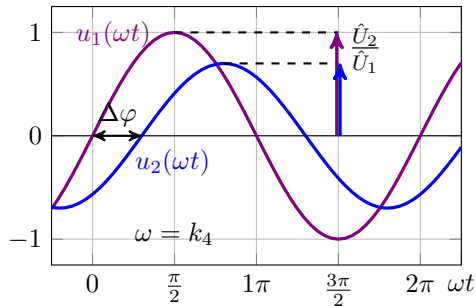


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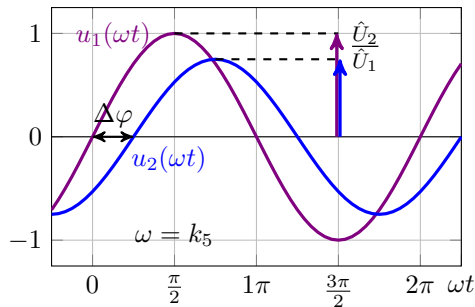


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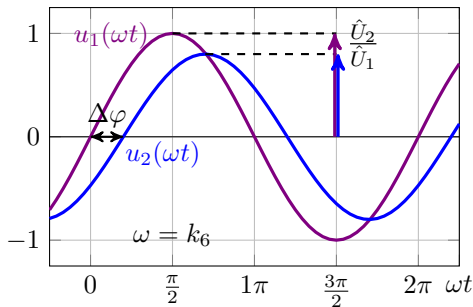


Analysis of frequency-variable behaviour of electrical networks (Two-port networks)



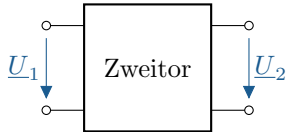
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Definition of frequency response, amplitude response and phase response

Zweitor (Vierpol):



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Linear, zeitinvariant (LZI):

$$u_1(t) = \hat{U}_1 \cdot \sin(\omega t)$$

$$\xrightarrow{LZI} u_2(t) = \hat{U}_2 \cdot \sin(\omega t + \varphi)$$

Sinusform bleibt erhalten

Frequenz gleich $\omega_2 = \omega_1$

Definition of frequency response, amplitude response and phase response

Zweitor (Vierpol):



Frequenzgang:

$$\underline{F}(j\omega) = \frac{\underline{U}_2}{\underline{U}_1} = A(\omega) \cdot e^{j\varphi(\omega)}$$

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Amplitudengang:

$$\begin{aligned} A(\omega) &= |\underline{F}(j\omega)| = \frac{|\underline{U}_2|}{|\underline{U}_1|} \\ &= \sqrt{(\operatorname{Re}\{\underline{F}(j\omega)\})^2 + (\operatorname{Im}\{\underline{F}(j\omega)\})^2} \end{aligned}$$

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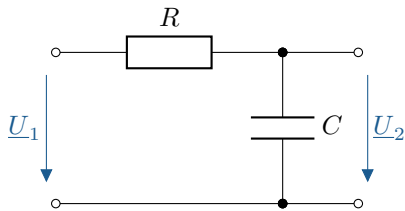
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Phasengang:

$$\begin{aligned} \varphi(\omega) &= \angle \underline{F}(j\omega) = \angle \underline{U}_2 - \angle \underline{U}_1 \\ &= \arctan \left(\frac{\operatorname{Im}\{\underline{F}(j\omega)\}}{\operatorname{Re}\{\underline{F}(j\omega)\}} \right) \end{aligned}$$

Frequency response using the example of a simple low-pass filter

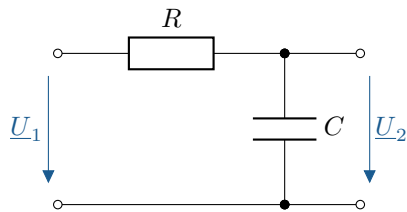


RC-Tiefpass 1. Ordnung

$$u_1(t) = \hat{U}_1 \cdot \sin(\omega t)$$

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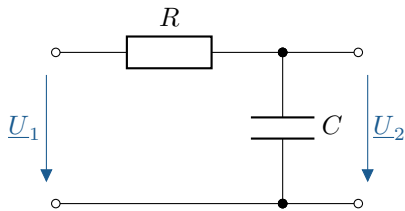
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Frequenzgang:

$$\begin{aligned} \underline{F}(j\omega) &= \frac{\underline{U}_2}{\underline{U}_1} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{j\omega CR + 1} \\ &= \frac{1}{1 + j\omega CR} = \frac{1 - j\omega CR}{1 + (\omega CR)^2} \end{aligned}$$

Frequency response using the example of a simple low-pass filter



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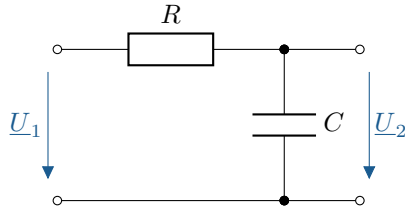
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Amplituden- und Phasengang:

$$A(\omega) = |\underline{F}(j\omega)| = \frac{|\underline{U}_2|}{|\underline{U}_1|} = \frac{1}{\sqrt{1 + (\omega CR)^2}}$$

$$\varphi(\omega) = \arctan\left(\frac{\text{Im}\{\underline{F}\}}{\text{Re}\{\underline{F}\}}\right) = \arctan(-\omega CR)$$

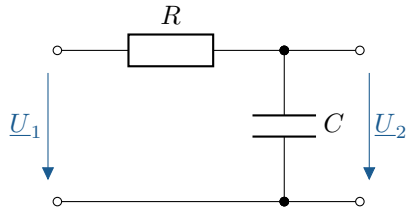
Boundary behaviour using the example of a simple low-pass filter



Grenzverhalten: für $f \rightarrow 0$ und $f \rightarrow \infty$

$$\underline{F}(j\omega) = A(\omega) \cdot e^{j\varphi(\omega)} = \frac{1}{1 + j\omega CR}$$

Boundary behaviour using the example of a simple low-pass filter

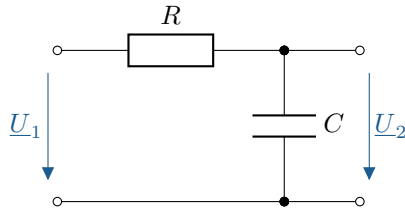


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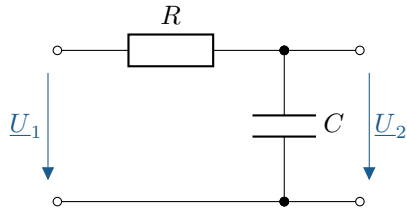
$$\underline{F}(j\omega) = A(\omega) \cdot e^{j\varphi(\omega)} = \frac{1}{1 + j\omega CR}$$

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega CR)^2}}$$

$$\lim_{\omega \rightarrow 0} A(\omega) = \frac{1}{1} = 1$$

$$\lim_{\omega \rightarrow \infty} A(\omega) = \frac{1}{\sqrt{\infty}} = 0$$

Boundary behaviour using the example of a simple low-pass filter



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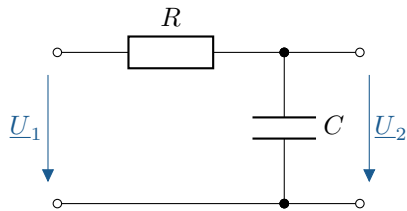
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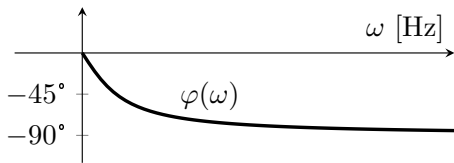
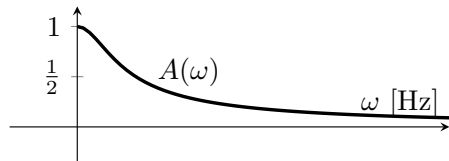
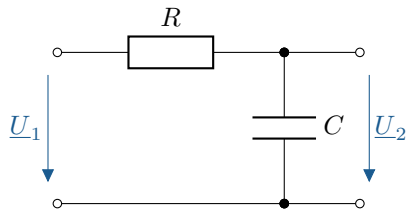
$$\lim_{\omega \rightarrow \infty} A(\omega) = \frac{1}{\sqrt{\infty}} = 0$$

$$\varphi(\omega) = \arctan(-\omega CR)$$

$$\lim_{\omega \rightarrow 0} \varphi(\omega) = \arctan(0) = 0^\circ$$

$$\lim_{\omega \rightarrow \infty} \varphi(\omega) = \arctan(-\infty) = -90^\circ$$

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$$A(\omega) = \frac{1}{\sqrt{1 + (\omega CR)^2}}$$

$$\lim_{\omega \rightarrow 0} A(\omega) = \frac{1}{1} = 1$$

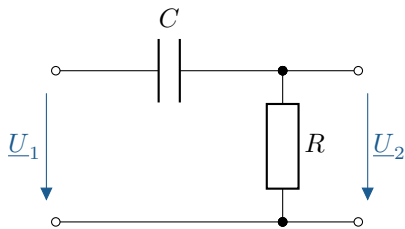
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Frequency response using the example of a simple high-pass filter



RC-Hochpass 1. Ordnung

$$u_1(t) = \hat{U}_1 \cdot \sin(\omega t)$$

$$\xrightarrow{LZI} u_2(t) = \hat{U}_2 \cdot \sin(\omega t + \varphi)$$

Frequency response:

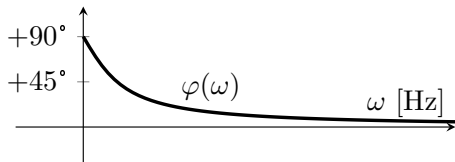
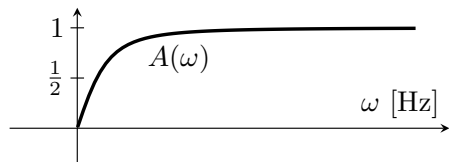
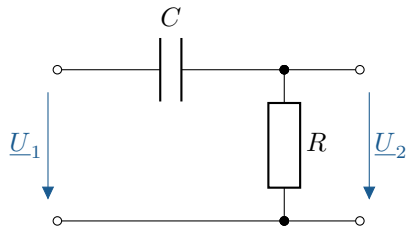
$$\begin{aligned} \underline{F}(j\omega) &= \frac{\underline{U}_2}{\underline{U}_1} = \frac{R}{R + \frac{1}{j\omega C}} = \frac{1}{1 + \frac{1}{j\omega CR}} \\ &= \frac{1}{1 - j \frac{1}{\omega CR}} = \frac{1 + j \frac{1}{\omega CR}}{1 + \frac{1}{(\omega CR)^2}} \end{aligned}$$

Amplitude and phase response:

$$A(\omega) = |\underline{F}(j\omega)| = \frac{|\underline{U}_2|}{|\underline{U}_1|} = \frac{1}{\sqrt{1 + \frac{1}{(\omega CR)^2}}}$$

$$\varphi(\omega) = \arctan\left(\frac{\operatorname{Im}\{\underline{F}\}}{\operatorname{Re}\{\underline{F}\}}\right) = \arctan\left(\frac{1}{\omega CR}\right)$$

Boundary behaviour using the example of a simple high-pass filter



Limit behaviour: für $f \rightarrow 0$ und $f \rightarrow \infty$

$$\underline{F}(j\omega) = A(\omega) \cdot e^{j\varphi(\omega)} = \frac{1}{1 - j\frac{1}{\omega CR}}$$

$$A(\omega) = \frac{1}{\sqrt{1 + \left(\frac{1}{\omega CR}\right)^2}}$$

$$\lim_{\omega \rightarrow 0} A(\omega) = \frac{1}{\infty} = 0$$

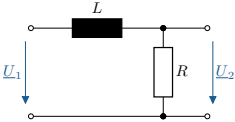
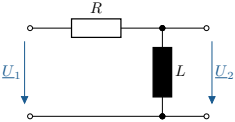
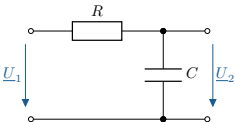
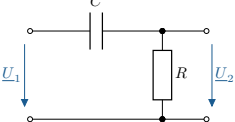
$$\lim_{\omega \rightarrow \infty} A(\omega) = \frac{1}{1} = 1$$

$$\varphi(\omega) = \arctan\left(\frac{1}{\omega CR}\right)$$

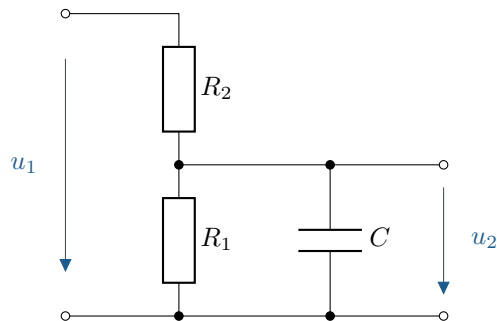
$$\lim_{\omega \rightarrow 0} \varphi(\omega) = \arctan(+\infty) = +90^\circ$$

$$\lim_{\omega \rightarrow \infty} \varphi(\omega) = \arctan(0) = 0^\circ$$

Comparison of first-order passive filters with L and R

	Low-pass filter	$\underline{F}(j\omega)$	High-pass filter	$\underline{F}(j\omega)$	ω_g
RL		$\frac{1}{1 + j\omega \frac{L}{R}}$		$\frac{1}{1 - j\frac{R}{\omega L}}$	$\frac{R}{L}$
RC		$\frac{1}{1 + j\omega CR}$		$\frac{1}{1 - j\frac{1}{\omega CR}}$	$\frac{1}{CR}$

Beispiel Filterschaltung: Aufgabe



Aufgabe:

- a) Leiten Sie $\underline{F}(j\omega) = \frac{\underline{U}_2}{\underline{U}_1}$ allgemein her.
- b) Zeichnen Sie Betrag und Phase von $\underline{F}(j\omega)$ für:

$$R_1 = 900 \, \Omega$$

$$R_2 = 100 \, \Omega$$

$$C = 1,25 \, \mu\text{F}$$

- c) Bei welcher Frequenz f ist $\hat{U}_2 = \frac{1}{2}\hat{U}_1$? Welche Phasenverschiebung besitzt u_2 nun gegenüber u_1 ?

$$\begin{aligned} u_1(t) &= \hat{U}_1 \cdot \sin(\omega t) & \underline{U}_1 &= \hat{U}_1 \\ u_2(t) &= \hat{U}_2 \cdot \sin(\omega t + \varphi) & \underline{U}_2 &= \hat{U}_2 \cdot e^{j\varphi} \end{aligned}$$

Solution a) Frequency response in general

$$\begin{aligned}\underline{F}(j\omega) &= \frac{R_1 || \underline{Z}_C}{R_2 + (R_1 || \underline{Z}_C)} = \frac{\frac{\frac{R_1}{j\omega C}}{R_1 + \frac{1}{j\omega C}}}{R_2 + \frac{\frac{R_1}{j\omega C}}{R_1 + \frac{1}{j\omega C}}} \left| \cdot \frac{R_1 + \frac{1}{j\omega C}}{R_1 + \frac{1}{j\omega C}} \right. \\&= \frac{\frac{R_1}{j\omega C}}{R_1 \cdot R_2 + \frac{R_2}{j\omega C} + \frac{R_1}{j\omega C}} = \frac{R_1}{R_1 + R_2 + j\omega C R_1 R_2} \quad \hat{=} \frac{a + jb}{c + jd} \quad a, b, c, d \in \mathbb{R} \\&= \frac{R_1(R_1 + R_2) - j\omega C R_1^2 R_2}{(R_1 + R_2)^2 + (\omega C R_1 R_2)^2} \quad (\text{kartesisch}) \quad \hat{=} \frac{a' + jb'}{c'} \quad a', b', c' \in \mathbb{R}\end{aligned}$$

$$\underline{F}(j\omega) = \frac{R_1}{\underbrace{R_1 + R_2 + j\omega C R_1 R_2}_{\hat{=} \frac{a+jb}{c+jd}}} = \frac{R_1(R_1 + R_2) - j\omega C R_1^2 R_2}{\underbrace{(R_1 + R_2)^2 + (\omega C R_1 R_2)^2}_{\hat{=} \frac{a'+jb'}{c'}}$$

Betrag: $|A(\omega)| = \frac{R_1}{\sqrt{(R_1 + R_2)^2 + (\omega C R_1 R_2)^2}}$ with $|\underline{F}(j\omega)| = \frac{\sqrt{a^2 + b^2}}{\sqrt{c^2 + d^2}}$

Phase: $\varphi(\omega) = \arctan\left(\frac{-\omega C R_1^2 R_2}{R_1(R_1 + R_2)}\right)$ with $\varphi(\omega) = \arctan\left(\frac{b'}{a'}\right)$

$$= \arctan\left(\frac{-\omega C R_1 R_2}{R_1 + R_2}\right)$$

Altern.: $\varphi(\omega) = \arctan(0) - \arctan\left(\frac{\omega C R_1 R_2}{R_1 + R_2}\right)$ with $\varphi(\omega) = \overbrace{\arctan(\frac{b}{a})}^{\varphi_Z} - \overbrace{\arctan(\frac{d}{c})}^{\varphi_N}$

Solution b) Limit behaviour and sketch

$$A(\omega) = \frac{R_1}{\sqrt{(R_1+R_2)^2 + (\omega C R_1 R_2)^2}}$$

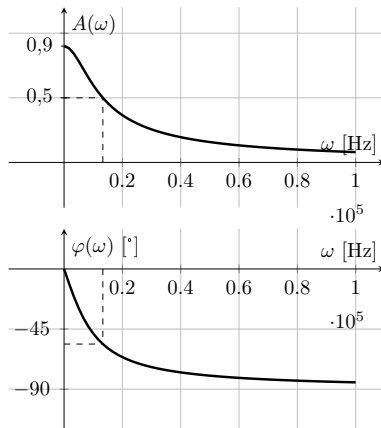
$$\lim_{\omega \rightarrow 0} A(\omega) = \frac{R_1}{R_1+R_2} = 0,9$$

$$\lim_{\omega \rightarrow \infty} A(\omega) = \frac{R_1}{\infty} = 0$$

$$\varphi(\omega) = \arctan\left(-\frac{\omega C R_1 R_2}{R_1+R_2}\right)$$

$$\lim_{\omega \rightarrow 0} \varphi(\omega) = \arctan(0) = 0$$

$$\lim_{\omega \rightarrow \infty} \varphi(\omega) = \arctan(-\infty) = -90^\circ$$



Solution c) Determine specific operating point

Total: Frequency f_1 where $\hat{U}_2 = \frac{1}{2}\hat{U}_1$ and phase shift u_2 relative to u_1 at f_1 :

$$A(\omega_1) = \frac{R_1}{\sqrt{(R_1 + R_2)^2 + (\omega CR_1R_2)^2}} \stackrel{!}{=} \frac{1}{2} \quad \text{with} \quad A(\omega) = \frac{U_2}{U_1} = \frac{\hat{U}_2}{\hat{U}_1} \stackrel{!}{=} \frac{1}{2} \Big|_{f=f_1}$$

$$\Rightarrow \frac{(R_1 + R_2)^2 + (\omega_1 CR_1R_2)^2}{R_1^2} = 4 \quad \Rightarrow \quad (R_1 + R_2)^2 + (\omega_1 CR_1R_2)^2 = 4R_1^2$$

$$\Rightarrow (\omega_1 CR_1R_2)^2 = 4R_1^2 - (R_1 + R_2)^2 \quad \Rightarrow \quad \omega_1 = \pm \frac{\sqrt{4R_1^2 - (R_1^2 + 2R_1R_2 + R_2^2)}}{CR_1R_2}$$

$$\omega_1 = 1,3304 \cdot 10^4 \frac{1}{s}, f_1 = 2,117 \text{ kHz} \quad \text{with} \quad \omega_1 < 0 \text{ unphysikalisch, } f = \frac{\omega}{2\pi}$$

$$\varphi(\omega_1) = \arctan(-1,4967) = -56,25^\circ \quad \text{with} \quad \varphi(\omega) = \arctan\left(\frac{\text{Im}\{\underline{F}\}}{\text{Re}\{\underline{F}\}}\right)$$

Learning objectives: Logarithmische Darstellung, Frequenzgang

Students learn:

- ▶ Know the differences between linear and logarithmic representation.
- ▶ Know and use decibels as a unit of measurement.
- ▶ Know and determine the cut-off frequencies of filter circuits.
- ▶ Construct and interpret Bode diagrams.
- ▶ Know and understand how bandpass and band-stop filters work.
- ▶ Know and understand the structure and behaviour of higher-order filters.

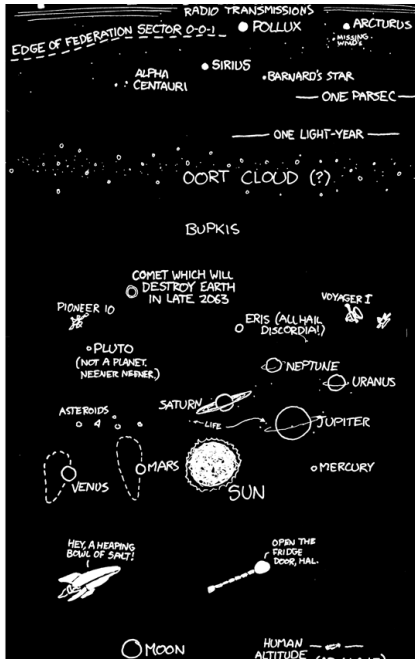
Logarithmic representation, frequency response, cut-off frequency



Multipage Comic Strip (vertikal Scrollen)

Quelle: xkcd #482, *Depth*, <https://xkcd.com/485/> (Abruf: 04.04.2024)

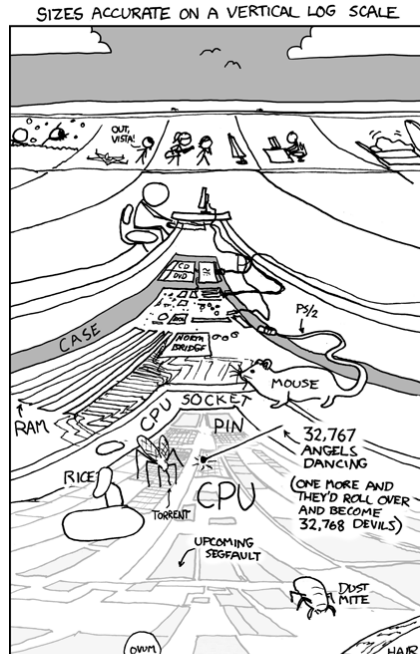
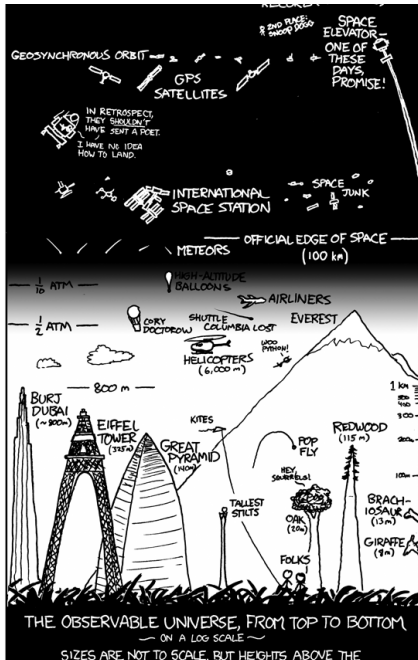
Logarithmic representation, frequency response, cut-off frequency



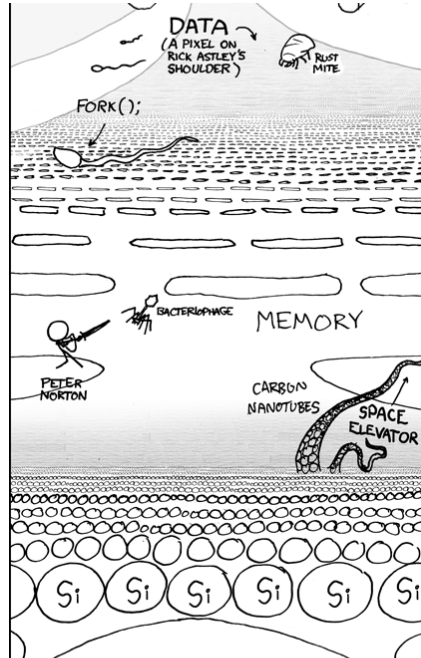
Multipage Comic Strip (vertikal Scrollen)

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Logarithmic representation, frequency response, cut-off frequency

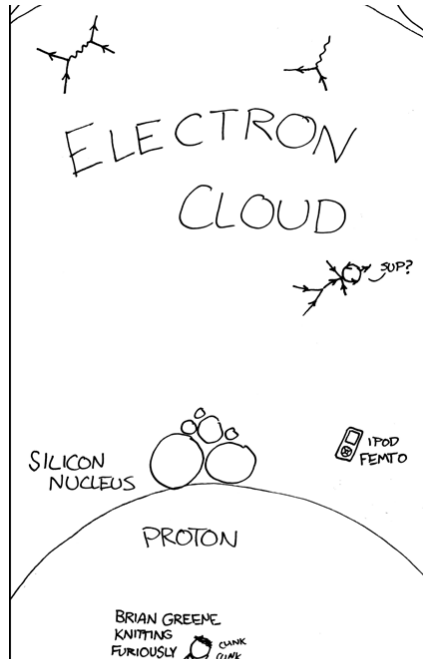


Logarithmic representation, frequency response, cut-off frequency



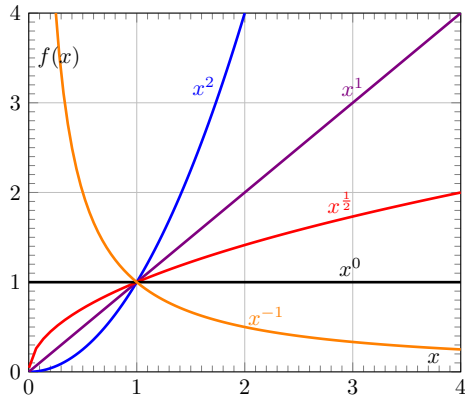
Quelle: xkcd #482, Height, <https://xkcd.com/482/> (Abruf: 04.04.2024)

Logarithmic representation, frequency response, cut-off frequency



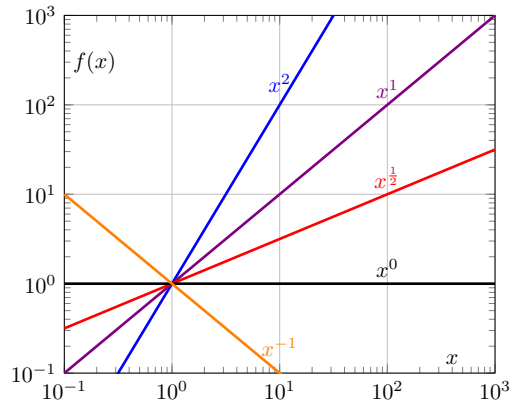
Quelle: xkcd #482, Height, <https://xkcd.com/482/> (Abruf: 04.04.2024)

Logarithmic representation of power functions



Lineare Darstellung

- Gerade: $f(x) = a(x - x_0) + y_0$
- Add./Subtr. → Verschiebung
- Mult./Div. → Streckung



(Doppelt-)Logarithmische Darstellung

- Gerade: $f(x) = (x/x_0)^a \cdot y_0$
- Mult./Div. → Verschiebung
- Exponent → Streckung

Decibel Auxiliary unit used to indicate logarithmic ratios between two quantities.

$$\text{Leistungsgrößen (z.B. } P, W) \quad Q_{(P)} = \log \frac{P_2}{P_1} \quad B = 10 \cdot \log \frac{P_1}{P_2} \text{ dB}$$

$$\text{Leistungswurzelgr. (z.B. } U, I) \quad Q_{(R)} = \log \frac{U_2^2}{U_1^2} \quad B = 20 \cdot \log \frac{U_1}{U_2} \text{ dB}$$

Typical decibel values (amplification in dB) at voltage ratios

$\frac{U_2}{U_1}$	$\frac{1}{\sqrt{2}}$	1	$\sqrt{2}$	2	$\sqrt{10}$	10	20	100
Q_{dB}	-3 dB	0 dB	+3 dB	+6 dB	10 dB	20 dB	26 dB	40 dB

Cut-off frequency: f_g (, ω_g)

$$\left. \frac{U_2}{U_1} \right|_{\omega=\omega_g} := \frac{1}{\sqrt{2}} \quad \leftrightarrow \quad A(\omega_g) := \frac{1}{\sqrt{2}}$$

Cut-off frequency: f_g (, ω_g)

$$\left. \frac{U_2}{U_1} \right|_{\omega=\omega_g} := \frac{1}{\sqrt{2}} \quad \leftrightarrow \quad A(\omega_g) := \frac{1}{\sqrt{2}}$$

Sections:

Durchlassbereich: $A(\omega) > 1/\sqrt{2}$

Grenzfrequenz: $A(\omega_g) = 1/\sqrt{2}$

Sperrbereich: $A(\omega) < 1/\sqrt{2}$

Cut-off frequency of simple filter circuits

Cut-off frequency: f_g (, ω_g)

$$\left. \frac{U_2}{U_1} \right|_{\omega=\omega_g} := \frac{1}{\sqrt{2}} \quad \leftrightarrow \quad A(\omega_g) := \frac{1}{\sqrt{2}}$$

Sections:

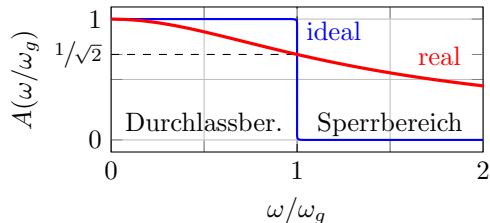
Durchlassbereich: $A(\omega) > 1/\sqrt{2}$

Grenzfrequenz: $A(\omega_g) = 1/\sqrt{2}$

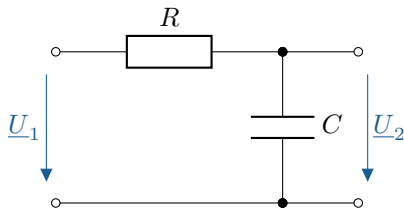
Sperrbereich: $A(\omega) < 1/\sqrt{2}$

Example: First-order low-pass filter

$$A(\omega/\omega_g) = \frac{1}{\sqrt{1 + (\omega/\omega_g)^2}} \quad \text{mit } \omega_g = \frac{1}{CR}$$



Logarithmische Darstellung



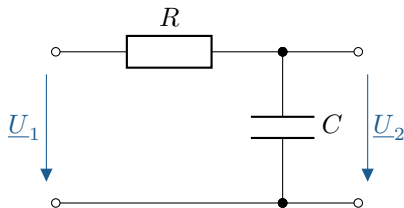
RC-Tiefpass 1. Ordnung

$$\underline{F}(j\omega) = \frac{1}{1 + j\omega RC}$$

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\varphi(\omega) = -\arctan(\omega RC)$$

Logarithmische Darstellung



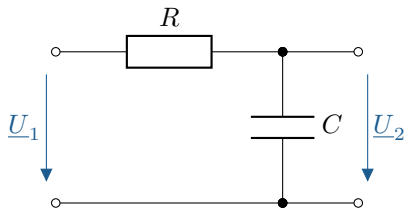
RC-Tiefpass 1. Ordnung

$$\underline{F}(j\omega) = \frac{1}{1 + j\omega RC}$$

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega RC)^2}} \quad A(\omega_g) = \frac{1}{\sqrt{2}}$$

$$\varphi(\omega) = -\arctan(\omega RC)$$

Logarithmische Darstellung



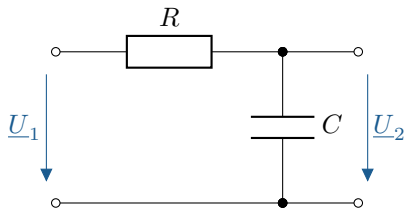
RC-Tiefpass 1. Ordnung

$$\underline{F}(j\omega) = \frac{1}{1 + j\omega RC} \quad \omega_g = \frac{1}{RC}$$

$$A(\omega) = \frac{1}{\sqrt{1 + (\omega RC)^2}} \quad A(\omega_g) = \frac{1}{\sqrt{2}}$$

$$\varphi(\omega) = -\arctan(\omega RC)$$

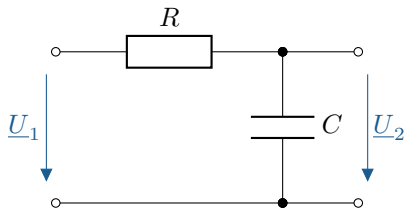
Logarithmische Darstellung



RC-Tiefpass 1. Ordnung

$$\begin{aligned}\underline{F}(j\omega) &= \frac{1}{1 + j\omega RC} & \omega_g &= \frac{1}{RC} \\ A(\omega) &= \frac{1}{\sqrt{1 + (\omega RC)^2}} & A(\omega_g) &= \frac{1}{\sqrt{2}} \\ \varphi(\omega) &= -\arctan(\omega RC) & \varphi(\omega_g) &= -45^\circ\end{aligned}$$

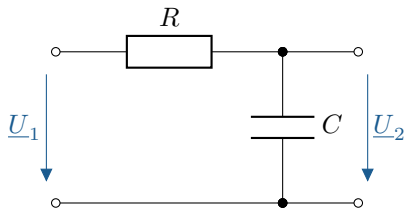
Logarithmische Darstellung



RC-Tiefpass 1. Ordnung

$$\begin{aligned}\underline{F}(j\omega) &= \frac{1}{1 + j\omega/\omega_g} & \omega_g &= \frac{1}{RC} \\ A(\omega) &= \frac{1}{\sqrt{1 + (\omega/\omega_g)^2}} & A(\omega_g) &= \frac{1}{\sqrt{2}} \\ \varphi(\omega) &= -\arctan(\omega/\omega_g) & \varphi(\omega_g) &= -45^\circ\end{aligned}$$

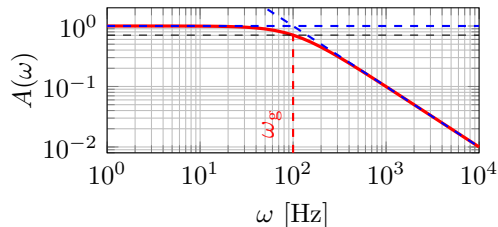
First-order low-pass filter



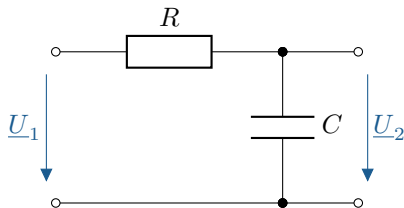
RC-Tiefpass 1. Ordnung

$$\begin{aligned}\underline{F}(j\omega) &= \frac{1}{1 + j\omega/\omega_g} & \omega_g &= \frac{1}{RC} \\ A(\omega) &= \frac{1}{\sqrt{1 + (\omega/\omega_g)^2}} & A(\omega_g) &= \frac{1}{\sqrt{2}} \\ \varphi(\omega) &= -\arctan(\omega/\omega_g) & \varphi(\omega_g) &= -45^\circ\end{aligned}$$

Logarithmische Darstellung



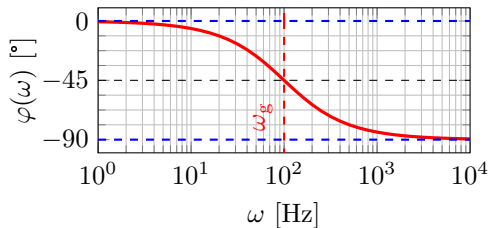
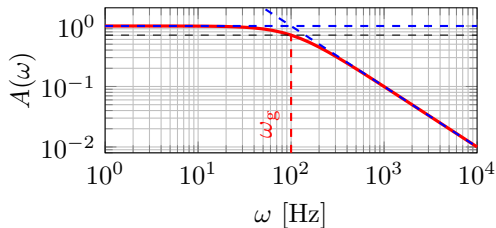
First-order low-pass filter



RC-Tiefpass 1. Ordnung

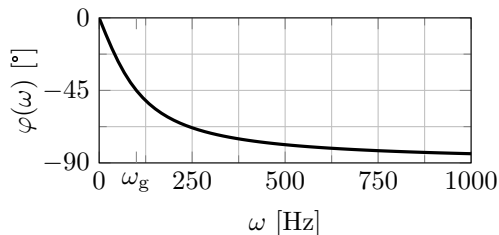
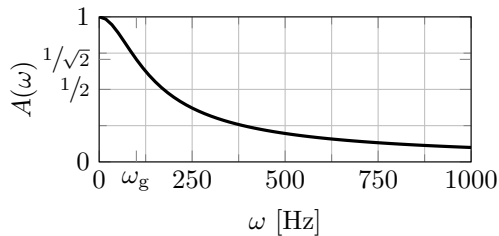
$$\begin{aligned} \underline{F}(j\omega) &= \frac{1}{1 + j\omega/\omega_g} & \omega_g &= \frac{1}{RC} \\ A(\omega) &= \frac{1}{\sqrt{1 + (\omega/\omega_g)^2}} & A(\omega_g) &= \frac{1}{\sqrt{2}} \\ \varphi(\omega) &= -\arctan(\omega/\omega_g) & \varphi(\omega_g) &= -45^\circ \end{aligned}$$

Logarithmische Darstellung

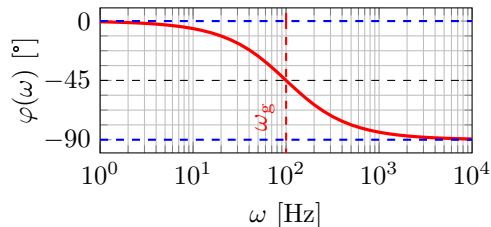
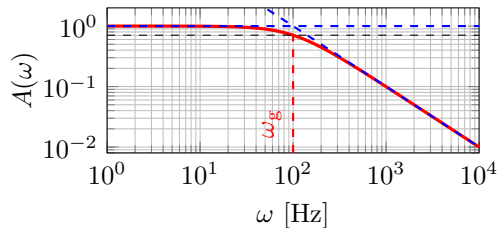


Comparison of linear and logarithmic representations

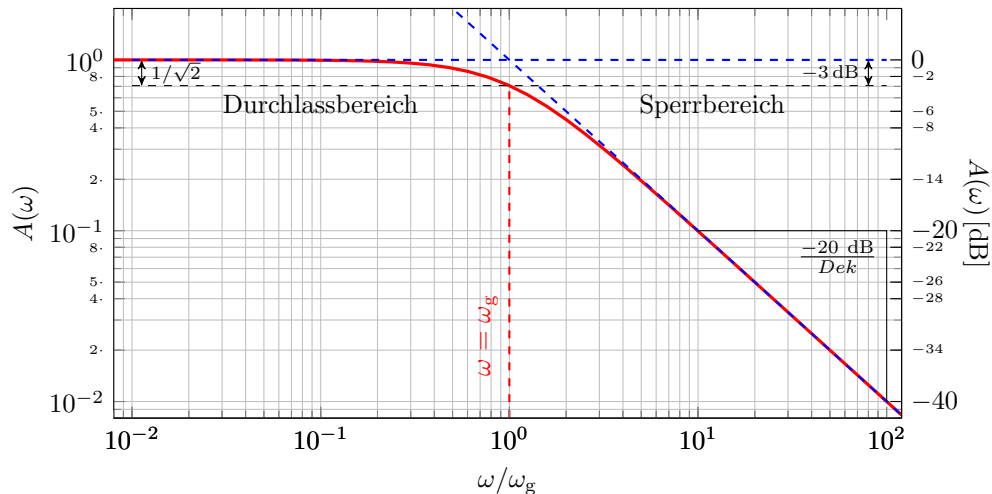
Lineare Darstellung



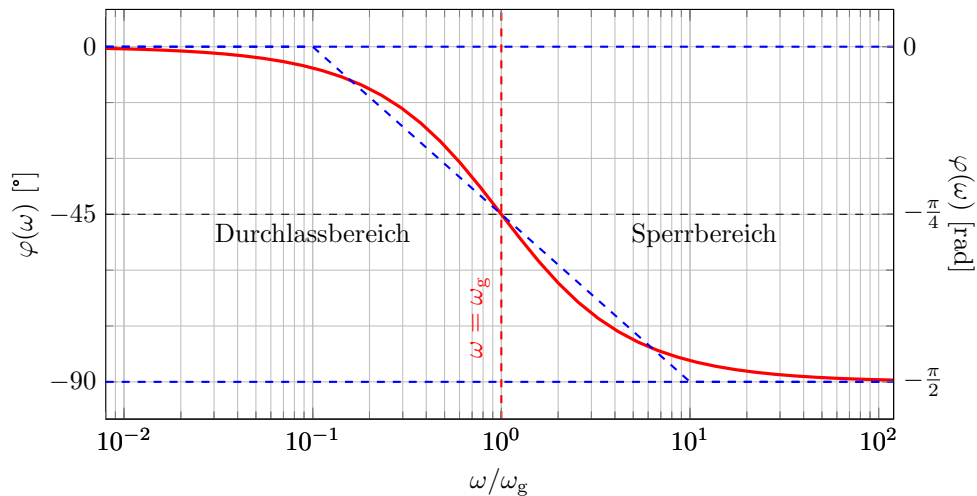
Logarithmische Darstellung



Tiefpass 1. Ordnung, Amplitudengang

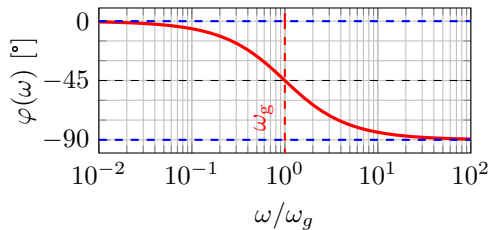
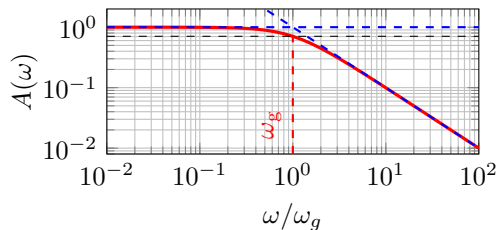


Tiefpass 1. Ordnung, Phasengang

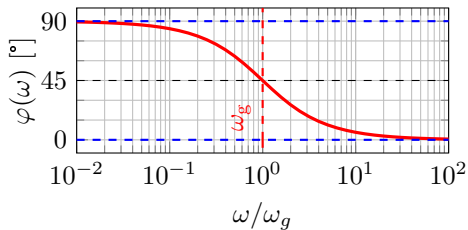
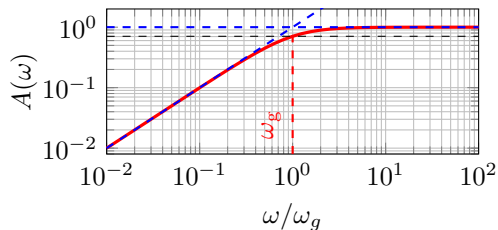


Comparison of first-order low-pass and high-pass filters

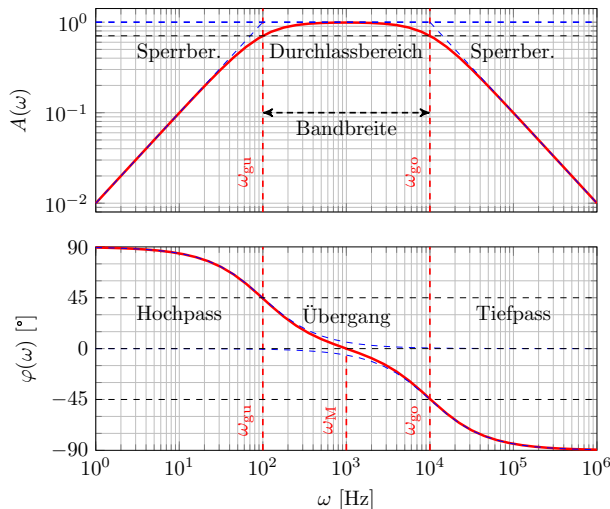
Tiefpass



Hochpass

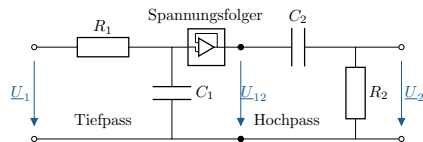


Second-order bandpass filter



Frequency response¹: $\underline{F}_{BP}(j\omega) = \underline{F}_{TP}(j\omega) \cdot \underline{F}_{TP}(j\omega)$

¹For non-reactive series connection from multiplication of individual frequency responses



Grenzfrequenz: (obere, untere)

$$f_{go}, f_{gu}$$

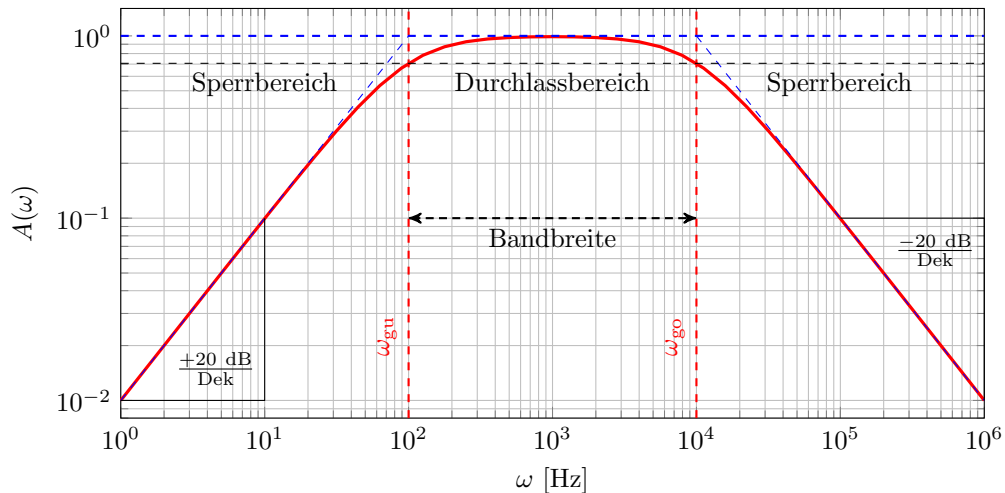
Bandbreite:

$$B = \Delta f = f_{go} - f_{gu}$$

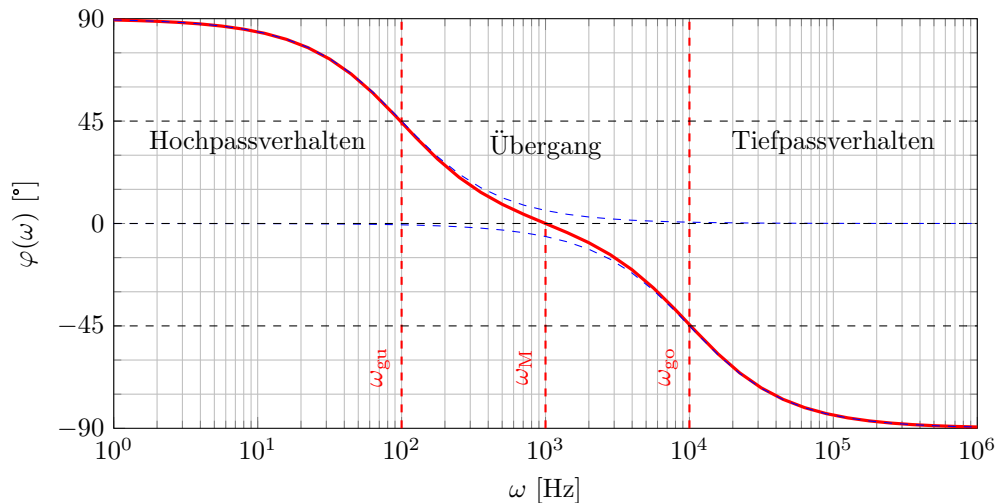
Mittenfrequenz:

$$f_m = \sqrt{f_{go} \cdot f_{gu}}$$

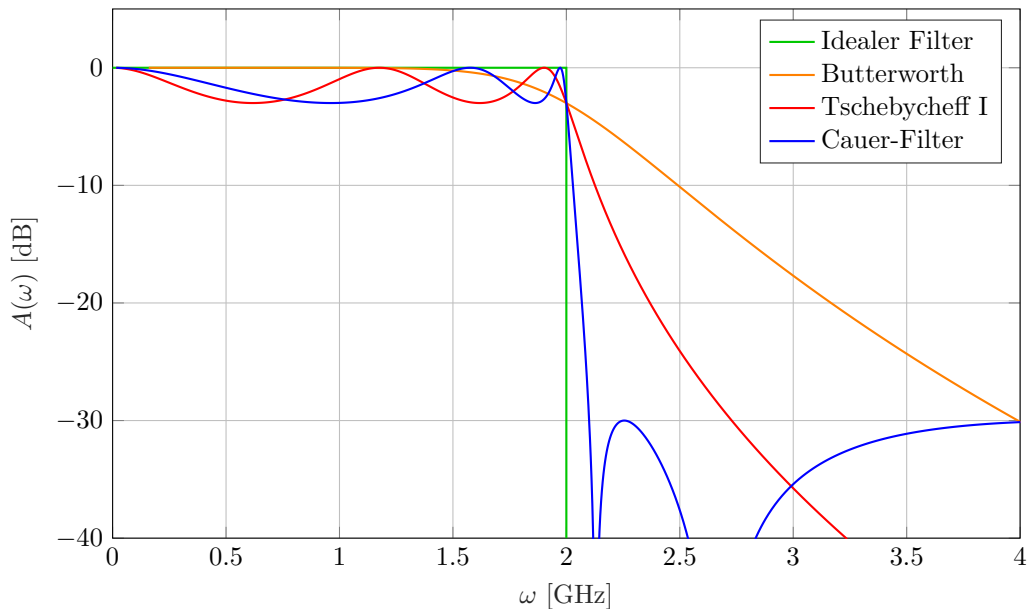
Bandpass 2. Ordnung - Amplitudengang, Detail



Bandpass 2. Ordnung - Phasengang, Detail



Order of filters, example filter of higher order



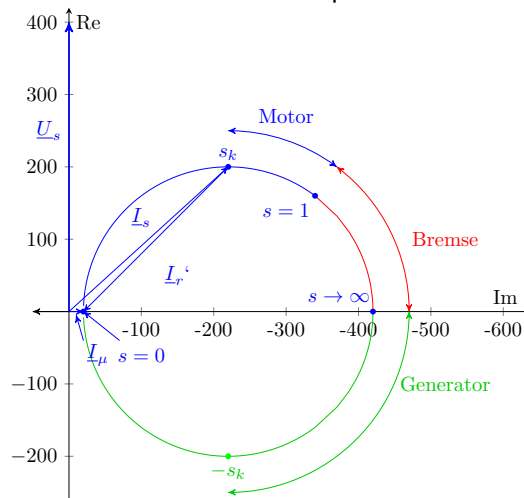
Learning objectives: Locus curve

Students learn:

- ▶ Know locus curves and possible areas of application.
- ▶ Construct and interpret locus curves in principle.
- ▶ Know impedance and admittance locus curves of basic circuits.

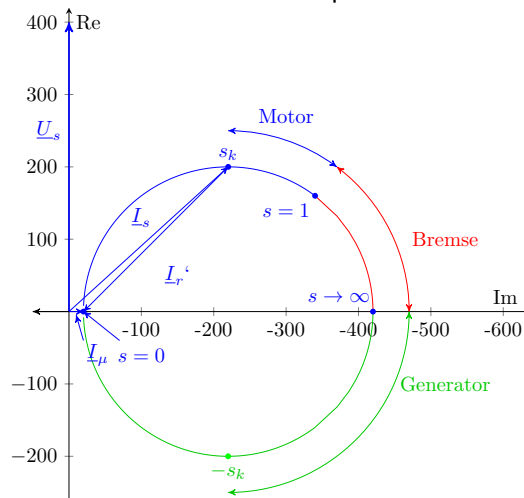
Def. Ortskurve: Darstellung eines komplexen Zeigers $\underline{z}(p) = \operatorname{Re}\{\underline{z}(p)\} + j \operatorname{Im}\{\underline{z}(p)\}$ in komplexer Ebene in Abhängigkeit eines realen Parameter p .

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Example: Stromortskurve einer Asynchronmaschine für $\underline{I}_s(s)$

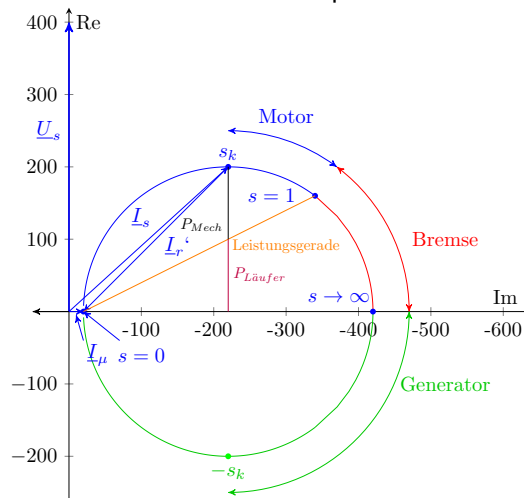
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Anwendungsbereiche?

Example: Stromortskurve einer Asynchronmaschine für $\underline{I}_s(s)$

Def. Ortskurve: Darstellung eines komplexen Zeigers $\underline{z}(p) = \operatorname{Re}\{\underline{z}(p)\} + j \operatorname{Im}\{\underline{z}(p)\}$ in komplexer Ebene in Abhängigkeit eines realen Parameter p .



Anwendungsbereiche?

Visualisierung von:

- Systemzuständen
- Systemgrößen
- Einfluss von Systemgrößen

Example: Stromortskurve einer Asynchronmaschine für $\underline{I}_s(s)$

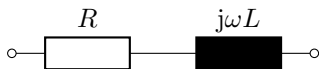
Zeigerdiagramm:

Zeigt Größe in komplexer Ebene als Zeiger (eines Punktes).

Key point: Location curve

Displays size in a complex plane as a curve (set of points) depending on parameters.

Impedanzortskurve - RL-Glied , R variabel

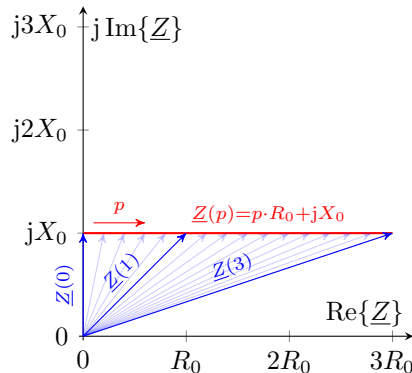


$$\underline{Z} = R + j\omega L$$

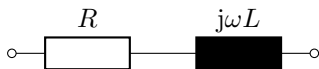
$$\underline{Z} = p \cdot R_0 + jX_0$$

mit $p \in [0,3]$

$R_0, X_0 = \text{konst.}$



Impedance locus curve – RLs element, ω variable

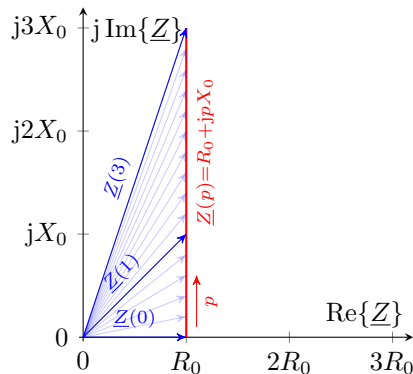


$$\underline{Z} = R + j\omega L$$

$$\underline{Z} = R_0 + jp \cdot \omega_0 L_0$$

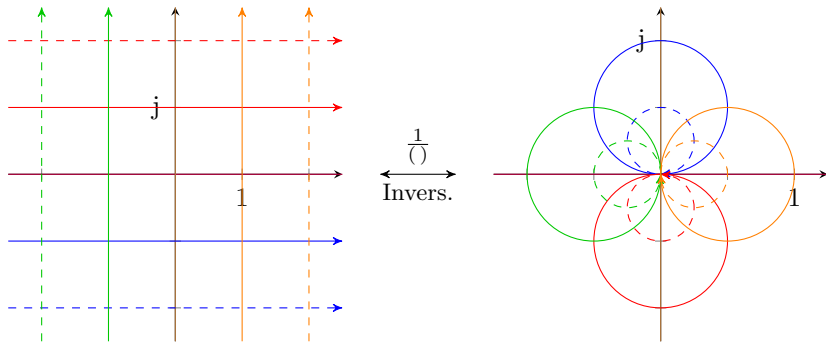
mit $p \in [0,3]$

$R_0, L_0, \omega_0 = \text{konst.}$



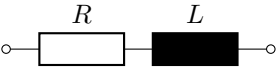
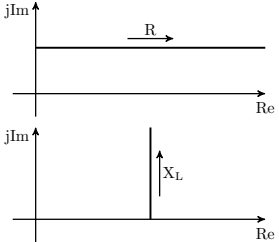
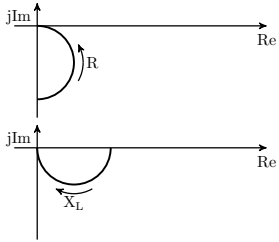
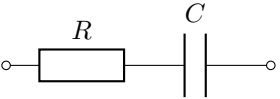
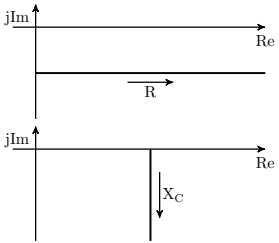
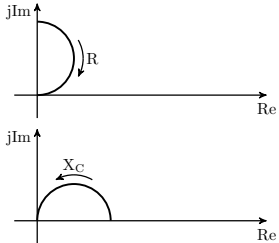
Admittance locus curve - Inversion of locus curves

original local curve		inverted locus curve	
Line	not through the origin	Line	not through the origin
	Not through the origin	Circle	through the origin
Circle	through the origin	Line	not through the origin
Circle	not through the origin	Circle	not through the origin

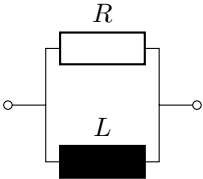
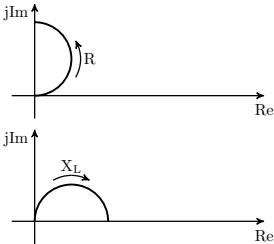
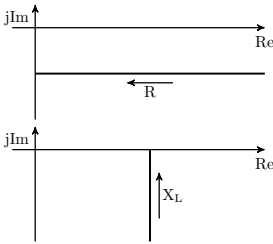
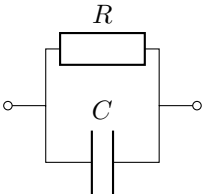
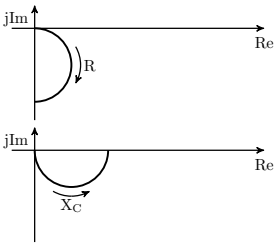
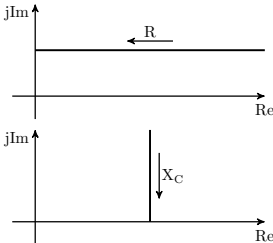


¹Based on MoebiusInversion.svg by Chrislb, CC-BY-SA-2.0-DE, 2005,
<https://commons.wikimedia.org/wiki/File:MoebiusInversion.svg>

Local curves of basic circuits - in Reihe

Basic-circuit	<u>Z</u> -Locus-curve	<u>Y</u> -Locus-curve
		
		

Local curves of basic circuits - in Parallel

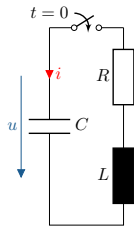
Grundschialtung	<u>Z</u> -Ortskurve	<u>Y</u> -Ortskurve
		
		

Learning objectives: Resonant circuits

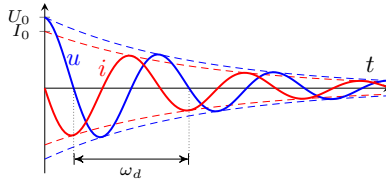
Students learn:

- ▶ Identifying and calculating resonance circuits
- ▶ Determine resonance frequency, characteristic impedance and quality factor
- ▶ Calculate, describe and plot resonance curves

Resonance phenomenon



Freie Schwingung:

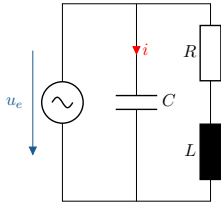


Randbedingung:

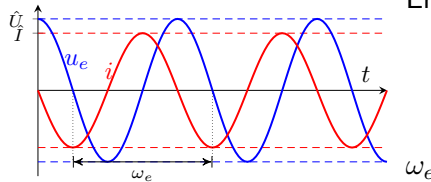
$$u(t = 0) = U_0$$

$$i(t = 0) = 0$$

mit Eigenkreisfrequenz ω_d ,
abklingend (gedämpft)



Forced oscillation:



Energizing voltage

$$u_e(t) = \hat{U} \sin(\omega_e t)$$

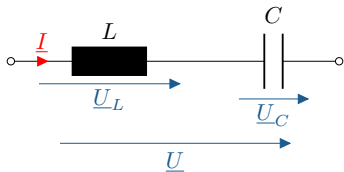
with Energizing frequency

ω_e

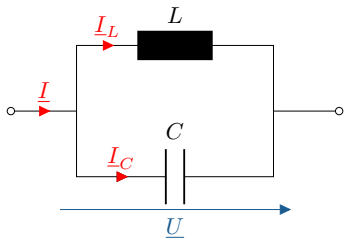
Definition of resonance condition, frequency, quality factor, characteristic impedance

Grundtypen

Reihen-Resonanzkreis



Parallel-Resonanzkreis



Resonanzbedingung $\text{Im}\{\underline{Z}\} \stackrel{!}{=} 0$

Resonanzfrequenz f_0

Kennwiderstand $X_k = |X_{L,0}| = |X_{C,0}|$

Kennleitwert $B_k = \frac{1}{X_k}$

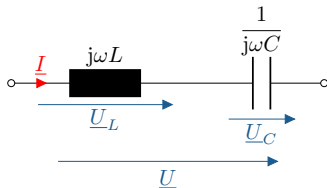
Güte $Q = \frac{|Q_{L,0}|}{P_0} = \frac{|Q_{C,0}|}{P_0}$

Eigenfrequenz f_d (gedämpft)

Eigenfrequenz f_0 (ungedämpft)

Resonant frequency using the example of ideal LC oscillating circuits

in Reihe

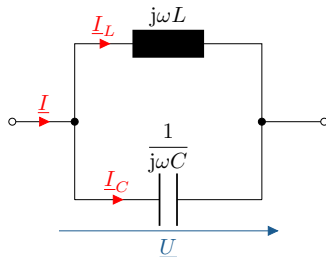


spannungsbezogen

$$\begin{aligned}\underline{U} &= \underline{I} \cdot \left(j\omega L + \frac{1}{j\omega C} \right) \\ &= \underline{I} \cdot j \left(\underbrace{\omega L - \frac{1}{\omega C}}_{=0 \text{ Resonanz}} \right)\end{aligned}$$

$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$

$$\Rightarrow \underline{I} \rightarrow \infty \quad \text{für} \quad \underline{U} = \text{konst.}$$



parallel

strombezogen

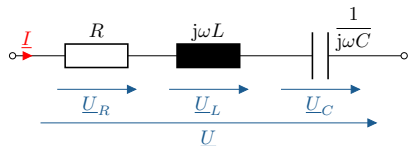
$$\begin{aligned}\underline{I} &= \underline{U} \cdot \left(j\omega C + \frac{1}{j\omega L} \right) \\ &= \underline{U} \cdot j \left(\underbrace{\omega C - \frac{1}{\omega L}}_{=0 \text{ Resonanz}} \right)\end{aligned}$$

$$\omega_0 C - \frac{1}{\omega_0 L} = 0$$

$$\Rightarrow \underline{U} \rightarrow \infty \quad \text{für} \quad \underline{I} = \text{konst.}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

RLC series resonant circuit - resonance behaviour



$$\underline{Z} = R + j \underbrace{\left(\omega L - \frac{1}{\omega C} \right)}_{= 0 \text{ für } \omega = \omega_0}$$

Resonance condition (allg.)

Phase shift between $\underline{U}$ und $\underline{I}$ entspricht $\varphi = 0$.

Or:

$$\operatorname{Im} \left\{ \frac{\underline{U}}{\underline{I}} \right\} = \operatorname{Im} \{ \underline{Z} \} \stackrel{!}{=} 0$$

Fall 1: $\omega < \omega_0$

$$\rightarrow \left(\omega L - \frac{1}{\omega C} \right) < 0 \quad \text{ohmsch-kapazitiv}$$

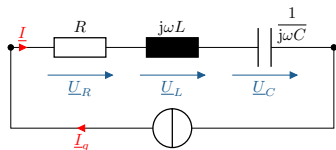
Fall 2: $\omega = \omega_0$ (Resonanzfall)

$$\rightarrow \left(\omega L - \frac{1}{\omega C} \right) = 0 \quad \text{rein ohmsch}$$

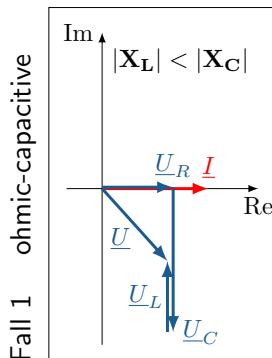
Fall 3: $\omega > \omega_0$

$$\rightarrow \left(\omega L - \frac{1}{\omega C} \right) > 0 \quad \text{ohmsch-induktiv}$$

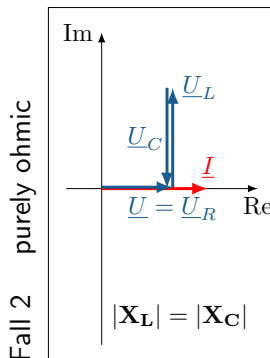
Case differentiation in the pointer diagram



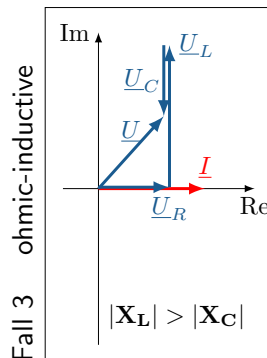
$$\begin{aligned}\underline{U} &= \underline{U}_R + \underline{U}_L + \underline{U}_C & \underline{I} &= \text{konst.} \\ &= \underline{I} \cdot \left[R + j \underbrace{\left(\omega L - \frac{1}{\omega C} \right)}_{= \operatorname{Im}\{Z\}} \right] & \omega_0 &= \frac{1}{\sqrt{LC}}\end{aligned}$$



$$\omega < \omega_0$$

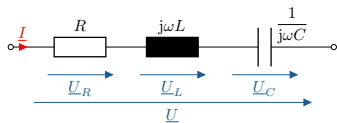


$$\omega = \omega_0$$



$$\omega > \omega_0$$

Case differentiation in impedance curve



$$\underline{Z} = R + j \left(\omega L - \frac{1}{\omega C} \right)$$

Resonanzbedingung

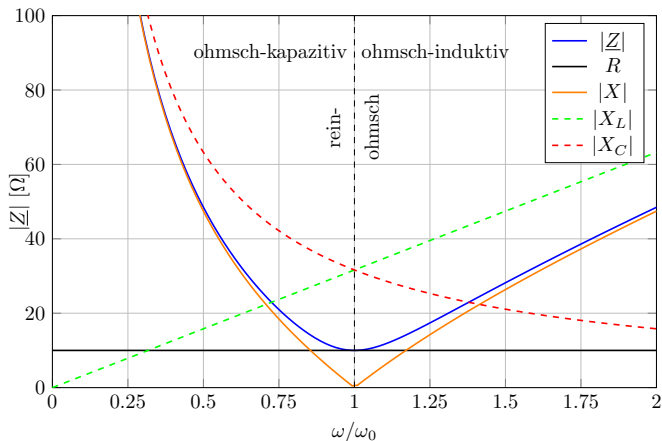
$$\text{Im}\{\underline{Z}\} \stackrel{!}{=} 0$$

Resonanzfrequenz

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

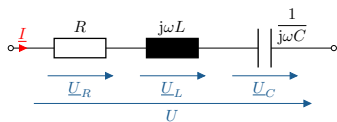
Resonanzimpedanz

$$\underline{Z}_0 = R$$



$$X_k = |X_{L,0}| = |X_{C,0}| = \sqrt{\frac{L}{C}}$$

Resonance curve and quality factor of an RLC series resonant circuit



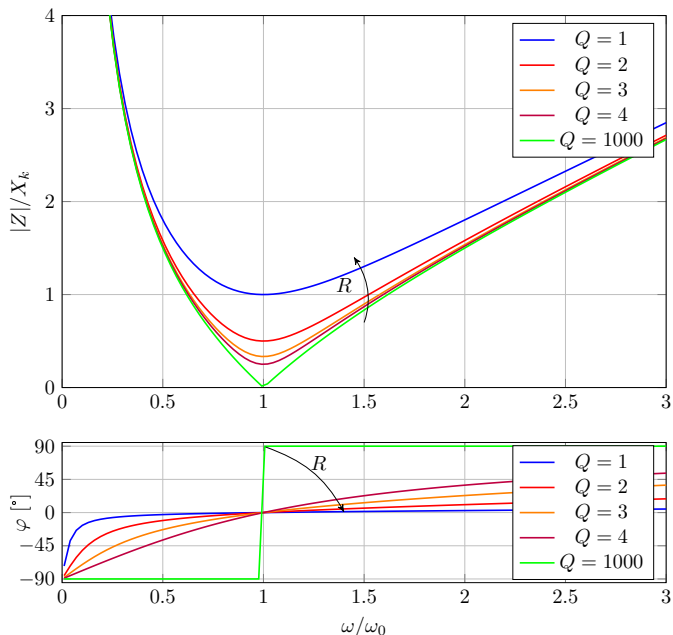
$$\underline{Z} = R + j \left(\omega L - \frac{1}{\omega C} \right)$$

normiert

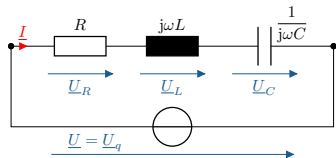
$$\frac{\underline{Z}}{X_k} = \frac{1}{Q} + j \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)$$

Güte und Kennwiderstand

$$Q = \frac{X_k}{R} \quad X_k = \sqrt{\frac{L}{C}}$$



Voltage resonance using the example of an RLC series resonant circuit I

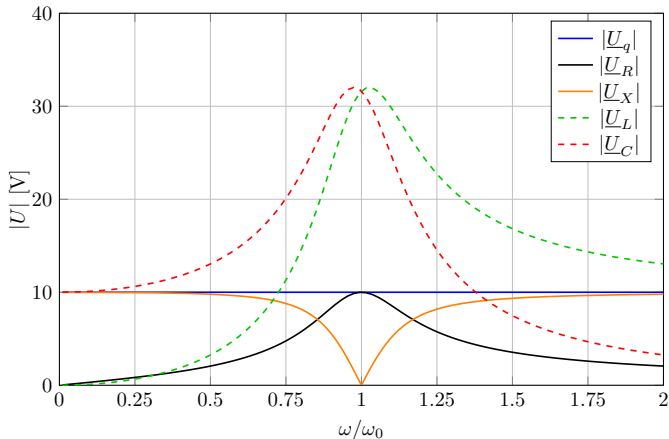


Spannungsteiler (Bsp.)

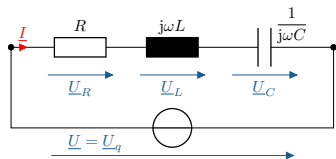
$$\frac{\underline{U}_L}{\underline{U}} = \frac{j \left(\frac{\omega}{\omega_0} \right)}{\frac{1}{Q} + j \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)}$$

Güte und Überspannung

$$Q = \frac{X_k}{R} = \frac{U_{L,0}}{U} = \frac{U_{C,0}}{U}$$



$U_q(\omega_0) \Rightarrow$ Überspannung!



$$\underline{U} = \underline{U}_R + \underline{U}_L + \underline{U}_C = \underline{U}_q$$

$$\underline{I} = \underline{I}_R = \underline{I}_L = \underline{I}_C$$

$$\underline{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\text{Im}\{\underline{Z}\} \stackrel{!}{=} 0 \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

$$X_k = |X_{L,0}| = |X_{C,0}| = \sqrt{\frac{L}{C}}$$

$$Q = \frac{U_{L,0}I}{U_{R,0}I} = \frac{U_{L,0}}{U_q} = \frac{X_k}{R}$$

$$\frac{\underline{U}_R}{\underline{U}} = \frac{\underline{I}}{\underline{I}} \cdot \frac{R}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

$$\frac{\underline{U}_L}{\underline{U}} = \frac{\underline{I}}{\underline{I}} \cdot \frac{j\omega L}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

$$\frac{\underline{U}_C}{\underline{U}} = \frac{\underline{I}}{\underline{I}} \cdot \frac{\frac{1}{j\omega C}}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

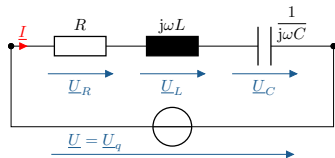
normiert

$$\frac{\underline{U}_R}{\underline{U}} = \frac{1/Q}{\frac{1}{Q} + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$

$$\frac{\underline{U}_L}{\underline{U}} = \frac{+j\frac{\omega}{\omega_0}}{\frac{1}{Q} + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$

$$\frac{\underline{U}_C}{\underline{U}} = \frac{-j\frac{\omega_0}{\omega}}{\frac{1}{Q} + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$

Current curve of the RLC series resonant circuit II

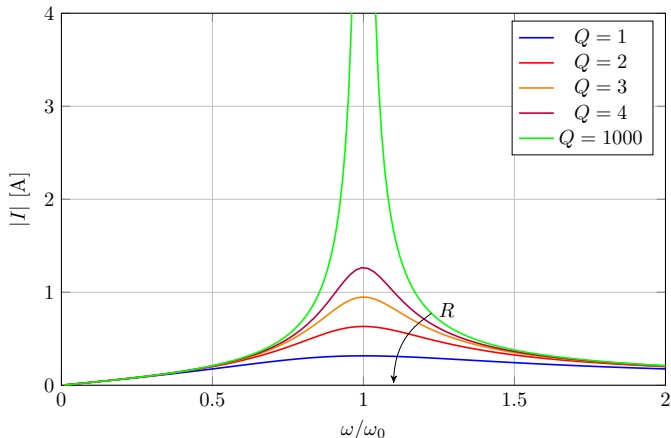


Strom

$$\underline{I} = \frac{I_0}{1 + j \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) Q}$$

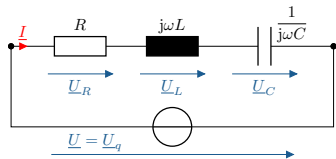
Maximalstrom, Güte

$$I_0 = \frac{U_q}{R} \quad Q = \frac{X_k}{R}$$



Strom maximal bei ω_0
Limitation by R

Additional calculation



$$\underline{Z} = R + j(X_L + X_C)$$

$$X_L = +X_k \frac{\omega}{\omega_0} = \omega L$$

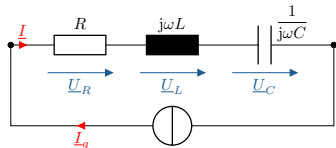
$$X_C = -X_k \frac{\omega_0}{\omega} = -\frac{1}{\omega C}$$

$$X_k = |X_{L,0}| = |X_{C,0}| = \sqrt{\frac{L}{C}}$$

$$Q = \frac{X_k}{R} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\begin{aligned} \underline{I} &= \frac{\underline{U}_q}{\underline{Z}} \\ &= \frac{\underline{U}_q}{R + j\left(\omega L - \frac{1}{\omega C}\right)} \\ &= \frac{\underline{U}_q}{R + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) X_k} \\ &= \frac{\underline{U}_q}{R} \cdot \frac{1}{1 + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) \frac{X_k}{R}} \\ &= \underline{I}_0 \cdot \frac{1}{1 + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) Q} \end{aligned}$$

Voltage curve of the RLC series resonant circuit



Stromquelle

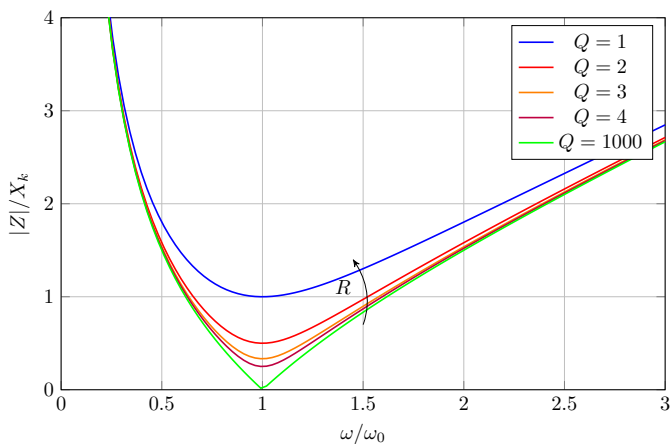
$$\underline{I} = I_q$$

Spannung

$$\underline{U} = I_q \cdot \underline{Z}$$

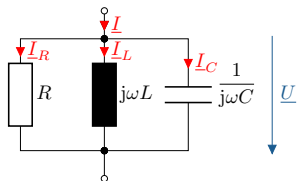
Impedanz

$$\underline{Z} = R + jX$$



Spannungskurve $\sim$ Impedanzkurve

RLC-Parallelschwingkreis - Resonanzverhalten



$$\underline{Y} = G + j \underbrace{\left(\omega C - \frac{1}{\omega L} \right)}_{= 0 \text{ für } \omega = \omega_0}$$

Resonanzbedingung (allg.)

Phasenverschiebung zwischen $\underline{U}$ und $\underline{I}$ entspricht $\varphi = 0$.

Oder:

$$\operatorname{Im} \left\{ \frac{\underline{I}}{\underline{U}} \right\} = \operatorname{Im} \{ \underline{Y} \} \stackrel{!}{=} 0$$

Fall 1: $\omega < \omega_0$

$$\rightarrow \left(\omega C - \frac{1}{\omega L} \right) < 0 \quad \text{ohmsch-induktiv}$$

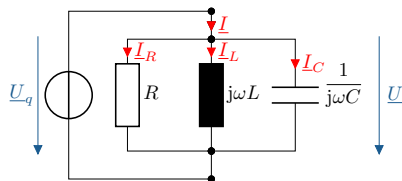
Fall 2: $\omega = \omega_0$ (Resonanzfall)

$$\rightarrow \left(\omega C - \frac{1}{\omega L} \right) = 0 \quad \text{rein ohmsch}$$

Fall 3: $\omega > \omega_0$

$$\rightarrow \left(\omega C - \frac{1}{\omega L} \right) > 0 \quad \text{ohmsch-kapazitiv}$$

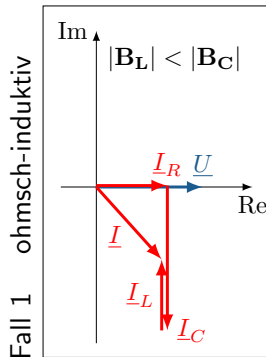
Case differentiation in the pointer diagram



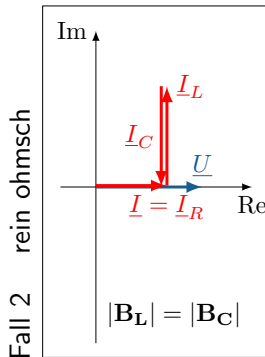
$$\begin{aligned}\underline{I} &= \underline{I}_R + \underline{I}_L + \underline{I}_C \\ &= \underline{U} \cdot \left[G + j \left(\omega C - \frac{1}{\omega L} \right) \right] \\ &= \underline{U} \cdot \underbrace{\left[G + j \left(\omega C - \frac{1}{\omega L} \right) \right]}_{= \text{Im}\{\underline{Y}\}}\end{aligned}$$

$$\underline{U} = \text{konst.}$$

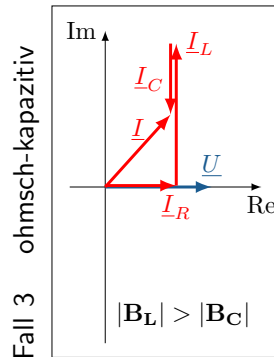
$$\omega_0 = \frac{1}{\sqrt{LC}}$$



$$\omega < \omega_0$$

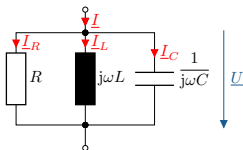


$$\omega = \omega_0$$



$$\omega > \omega_0$$

Case differentiation in admittance curve



$$\underline{Y} = G + j \left(\omega C - \frac{1}{\omega L} \right)$$

Resonanzbedingung

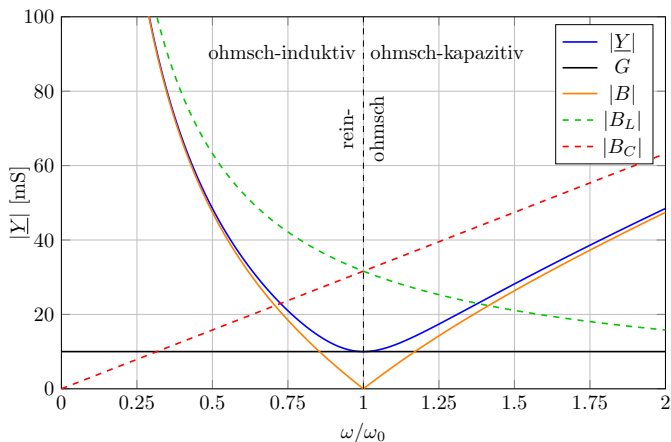
$$\text{Im}\{\underline{Y}\} \stackrel{!}{=} 0$$

Resonanzfrequenz

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

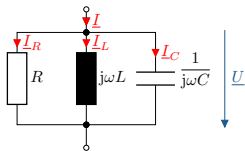
Resonanzadmittanz

$$\underline{Y}_0 = G$$



$$B_k = |B_{C,0}| = |B_{L,0}| = \sqrt{\frac{C}{L}}$$

Resonance curve and quality factor of an RLC parallel resonant circuit



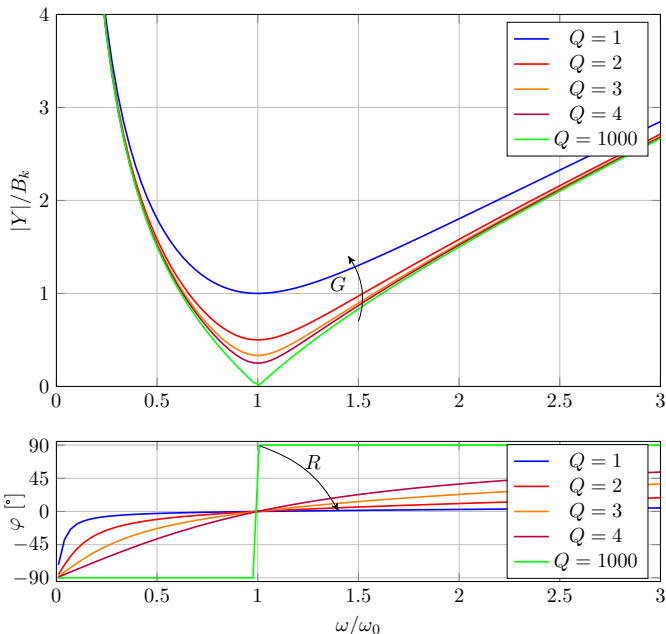
$$\underline{Y} = G + j \left(\omega C - \frac{1}{\omega L} \right)$$

normiert

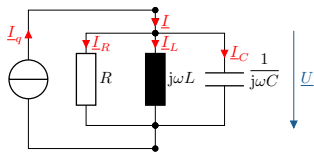
$$\frac{\underline{Y}}{B_k} = \frac{1}{Q} + j \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)$$

Güte und Kennleitwert

$$Q = \frac{B_k}{G} \quad B_k = \sqrt{\frac{C}{L}}$$



Current resonance using the example of an RLC parallel resonant circuit



$$\underline{U} = \underline{U}_R = \underline{U}_L = \underline{U}_C$$

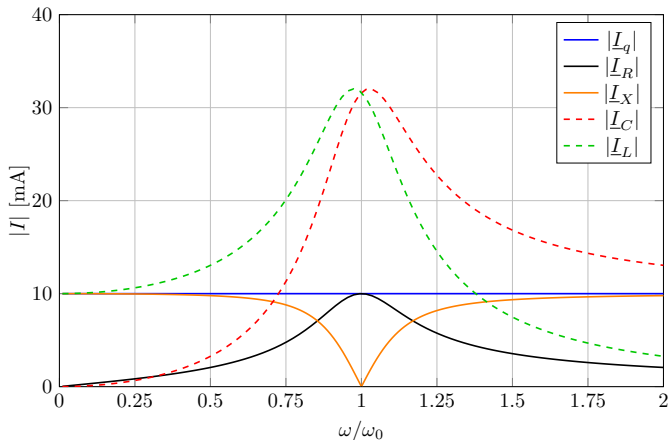
$$\underline{I} = \underline{I}_R + \underline{I}_L + \underline{I}_C = \underline{I}_q$$

Stromteiler (Bsp.)

$$\frac{\underline{I}_C}{\underline{I}} = \frac{j \left(\frac{\omega}{\omega_0} \right)}{\frac{1}{Q} + j \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)}$$

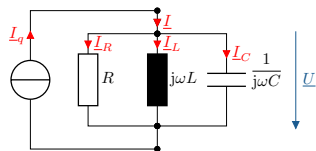
Güte und Überstrom

$$Q = \frac{B_k}{G} = \frac{I_{C,0}}{I} = \frac{I_{L,0}}{I}$$



Stromquelle $I_q(\omega_0) \Rightarrow$ Überstrom!

Voltage curve of the RLC parallel resonant circuit II

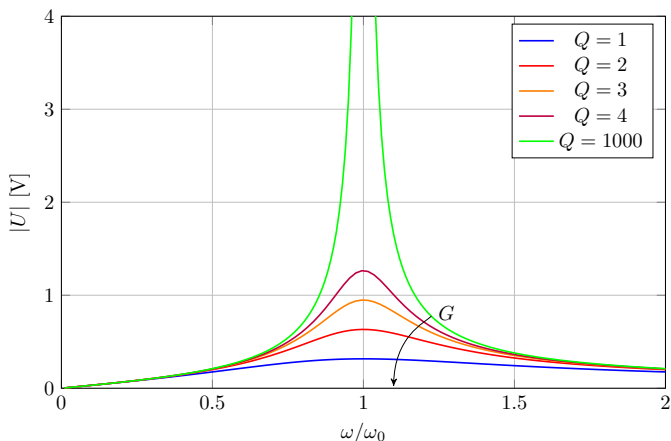


Spannung

$$\underline{U} = \frac{U_0}{1 + j \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right) Q}$$

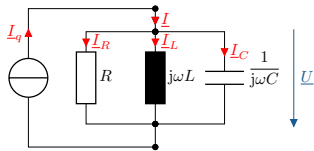
Maximalspannung, Güte

$$U_0 = \frac{I_q}{G} \quad Q = \frac{B_k}{G}$$



Spannung maximal bei ω_0
Begrenzung durch G

Additional calculation



$$\underline{Y} = G + j(B_L + B_C)$$

$$B_L = -B_k \frac{\omega_0}{\omega} = -\frac{1}{\omega L}$$

$$B_C = +B_k \frac{\omega}{\omega_0} = \omega C$$

$$B_k = |B_{L,0}| = |X_{C,0}| = \sqrt{\frac{C}{L}}$$

$$Q = \frac{B_k}{G} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\underline{U} = \frac{\underline{I}_q}{\underline{Y}}$$

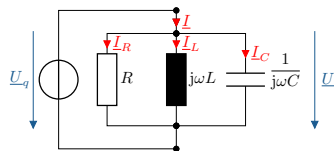
$$= \frac{\underline{I}_q}{G + j\left(\omega C - \frac{1}{\omega L}\right)}$$

$$= \frac{\underline{I}_q}{G + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) B_k}$$

$$= \frac{\underline{I}_q}{G} \cdot \frac{1}{1 + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) \frac{B_k}{G}}$$

$$= \underline{U}_0 \cdot \frac{1}{1 + j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) Q}$$

Current curve of the RLC parallel resonant circuit



Spannungsquelle

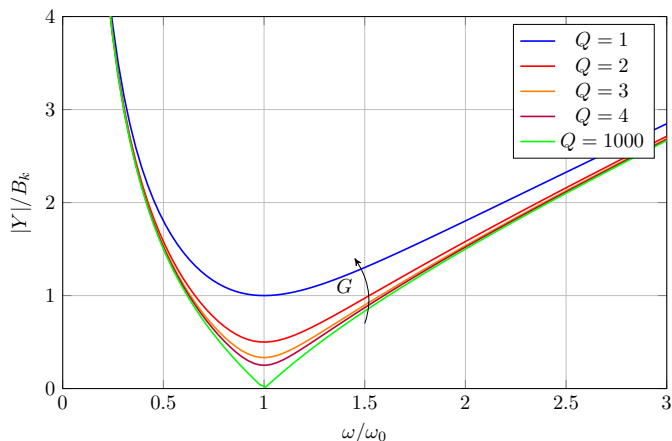
$$\underline{U} = U_q$$

Strom

$$\underline{I} = U_q \cdot \underline{Y}$$

Admittanz

$$\underline{Y} = G + jB$$



Stromkurve $\sim$ Admittanzkurve

